AquaDrone – steering tuning sheet

ArduRover 4.6.3, vectored thrust. Companion to docs/steering-control-analysis.md. Apply one step at a time, test, then continue. Save the current parameter file first.

Diagnosis in one breath

• Straight line: 0.43 Hz yaw limit cycle driven by P (±10 deg/s at 3 m/s, growing at 3.15 m/s).

• Slow 45–50 s cross-track weave: under-damped position loop (PSC_VEL_P too low).

• Corners: planner demands ~37 deg/s, boat is limited to 18 → servo on the stop, 5–6 m overshoot.

• Your instincts (no D, more I, less P) are right; the two items above are the other half.

Step 1 – yaw-rate loop (straight-line nervousness)

Kill the 0.43 Hz wobble. Loop has ~1 s dead time, so lower gain and let the integrator do the trim.

ATC_STR_RAT_P 0.35 → **0.15**	Limit cycle at 3 m/s means loop gain ≈ 1. Halving-plus gives > 6 dB margin.
ATC_STR_RAT_I 0.02 → **0.10**	Integral time P/I ≈ 1.5 s ≈ dead time. Removes the standing 1–2 deg/s error.
ATC_STR_RAT_IMAX 0.10 → **0.25**	Trim alone needs 0.02–0.05 output; room for wind/current.
ATC_STR_RAT_FF 0.60 → **0.50**	Rate overshoots 18 → 22 deg/s with P ≈ 0, so FF is high. 0.5 no longer saturates the mixer at 26 % throttle.
ATC_STR_RAT_D 0 → 0 (keep)	Dead-time plant: D adds noise, recovers no phase.
ATC_STR_RAT_SMAX 9 → **0**	Never active at this output scale; disable.
ATC_STR_RAT_FLTT / FLTE 2 / 2 → keep	Optional later: 4 / 4 gains ~6° phase, more gyro noise.
ATC_STR_RAT_MAX / ACC_MAX 18 / 6.5 → keep	18 deg/s ≈ 80 % of measured 22 deg/s capability.

Pass: ≥ 60 s straight at 2.2 and at 3.0 m/s: no 0.3–0.5 Hz yaw wobble (band rms < 1.5 deg/s), steering servo peak-to-peak < 100 µs, PID_TUNING shows P and I similar size and I settling.

If not: Wobble still visible at top speed → P 0.10. Rate lags demand > 2 s entering a corner → FF +0.05 steps.

Step 2 – position loop (slow 45–50 s weave)

Add damping to the cross-track loop (currently ζ ≈ 0.41, 51 s period).

PSC_VEL_P 0.10 → **0.40**	ζ ≈ 0.82, 26 s period. If it looks twitchy use 0.30 (ζ 0.71).
PSC_POS_P 0.15 → keep	Keeps outer loop ≥ 10× slower than rate loop.
PSC_VEL_D 0.02 → keep	

Pass: After a corner the cross-track error is back below 2 m with at most one overshoot; no sustained 40–50 s swing on straights.

If not: Overshoot still > 1 cycle → PSC_VEL_P 0.30. Sluggish return → PSC_POS_P 0.20.

Step 3 – corner planning (sharp corners)

Planned corner rate ≈ 2·WP_ACCEL / WP_SPEED. Today that is 37 deg/s against an 18 deg/s limit. Target ≈ 13 deg/s at the lowest WP_SPEED.

WP_ACCEL 0.8 → **0.25**	13 deg/s at 2.2 m/s, 9.5 at 3.0. Corner radius ≈ 7 m. Costs ~9 s / 10 m to reach 2.2 m/s from stop. Use 0.3 if you never run 2.2 m/s.
WP_JERK 0.5 → **0.25**	Rate onset ≈ 6.5 deg/s², matches ATC_STR_ACC_MAX.
ATC_TURN_MAX_G 0.6 → keep	Do NOT lower: below √2·WP_ACCEL/g the corner blend is delayed and corners get sharper.
TURN_RADIUS / WP_RADIUS 25 / 10 → keep	Only bound the apex distance; inactive here.

Pass: 90° corner: demanded rate < 15 deg/s and never clamped, servo not at 1200/1800 µs for > 0.5 s, cross-track after corner < 3 m.

If not: Still clamped → WP_ACCEL 0.2. Corners too wide/slow → WP_ACCEL 0.3.

Step 4 – speed loop & instrumentation

Make the throttle hunting diagnosable before touching ATC_SPEED_*.

GCS_PID_MASK 1 → **3**	Streams the speed PID (PID_TUNING axis 4).
CRUISE_SPEED / CRUISE_THROTTLE 1.30 / 31 (file) → **re-learn**	Log shows ≈ 2.3 m/s @ 26 %. Use CRUISE_LEARN aux switch at WP_SPEED, or read off a steady MANUAL run.
SRn_EXTRA1 / SRn_RC_CHAN 4 / 2 Hz → **10 / 10 Hz**	For tuning sessions only; or pull the on-board .bin log (STER, PIDS, PSCN/PSCE at 50 Hz).
ATC_SPEED_P / I 0.1 / 0.02 → keep for now	Decide with the speed-PID stream in hand.

Pass: Throttle no longer swings 13–36 % on straights; speed error < 0.3 m/s rms.

Hardware / config checks

• MOT_VEC_ANGLEMAX = 35 must equal the real thruster angle at SERVO1 1200 / 1800 µs.

• The ~1 s command-to-yaw delay is the hard limit on gain. A faster steering actuator is the only way to allow higher P and tighter corners.

• The saved .param file differs from the logged configuration (AHRS_ORIENTATION, COMPASS_USE2, cruise pair, WP_SPEED). Verify you are tuning the boat you logged.

Quick copy list

Only the values that change. Paste into Mission Planner / QGC one step at a time.

Step 1

```
ATC_STR_RAT_P,0.15
ATC_STR_RAT_I,0.10
ATC_STR_RAT_IMAX,0.25
ATC_STR_RAT_FF,0.50
ATC_STR_RAT_SMAX,0
```

Step 2

```
PSC_VEL_P,0.40
```

Step 3

```
WP_ACCEL,0.25
WP_JERK,0.25
```

Step 4

```
GCS_PID_MASK,3
```

Re-learn CRUISE_SPEED / CRUISE_THROTTLE at WP_SPEED before changing ATC_SPEED_*.